MindSync: Enhancing Mental Health and Well-being for College Students

A PROJECT REPORT

Submitted in partial fulfillment for the Degree of
Bachelor of Technology under the
School of Computing

Submitted by

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BONAFIDE CERTIFICATE

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Contents

Acknowledgments 5

List of Figures 6

List of Tables 7

Abstract 8

1 Introduction 9

1.1 Background

1.2 Problem Statement

1.3 Objective

1.4 Scope of the Study/ Project 10

2 Literature Review 11

2.1 Literature Review

2.1.1 Key Themes and Area Covered

2.1.2 Search Strategy 14

2.1.3 Insight from Prior Work

3 Proposed Work 16

3.1 Proposed Work

3.1.1 Objective of the Proposed Work 17

3.2 Methodology

3.2.1 Overview of the Approach

3.2.2 Dataset Selection 18

3.2.3 Algorithm

3.2.4 Tools and Technologies 19

3.2.5 System Architecture/Design

3.2.6 Algorithm Description

3.2.7 Expected Outcome 20

3.2.8 Advantages of the Proposed Work

3.2.9 Limitations and Assumptions

4 Experimentation and Results 21

4.1 Experimental Setup
4.2 Dataset
4.3 EvaluationMetrics
4.4 Experimental Design 23

4.4.1 Experimental Scenarios
4.4.2 Parameter Tuning 24

4.5 Results

4.6 Analysis of Results 25

4.6.1 Random Forest Model
4.6.2 Decision Tree

4.7 Observations 26

4.7.1 Random Forest Model
4.7.2 Decision Tree
4.7.3 Dataset Constraints
4.7.4 Algorithms Efficiency

4.8 Comparative Analysis

4.8.1 Limitations of the Proposed Architectures 27

5 Conclusion and Scope for Further Research 28

A Source Code for Mental Health Prediction

References

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List of Figures

3.1 System Architecture for Mental Health Prediction

4.1 Dataset collection format

A.1 Preprocessing

A.2 Training Random Forest Model

A.3 Prediction

List of Tables

2.1: Summary of Research on Mental Health Prediction

4.1: Model Comparison

Abstract

The field of mental health of college students is especially significant because students are going through great development during this such as learning, starting new social engagements, and struggles with stress, which can result in a negative impact on their health. There is a need to combat the negative impact of mental health in this category as it promotes resilience, better educational performance, and improved well-being. This project aims at the provision of effective and individualized mental health support using new technologies that will respond to the needs of students. Using website and machine learning models for data analytics, we expect to better understand students' mental health challenges and coping strategies. The results of this study may help to develop new methods of primary prevention, case management, and policy changes which would result in a decrease in mental health challenges and degree attainment of college students.

Chapter 1

Introduction

1.1 Background

Mental health challenges are prevalent among college students, driven by academic pressures, social transitions, and financial stressors. These factors often lead to increased rates of stress, anxiety, and depression, negatively impacting students' well-being and academic performance. Traditional support systems face barriers such as stigma and limited accessibility, creating a need for innovative, scalable solutions.

Digital tools, including websites and machine learning-driven interventions, offer a promising approach to addressing these challenges by providing accessible, personalized, and real-time support. This project explores the potential of such technologies to enhance mental health outcomes for students, fostering resilience, academic success, and overall well-being.

1.2 Problem Statement

Mental health challenges among college students are increasingly prevalent due to academic pressures social transitions and lifestyle changes. These factors contribute to stress anxiety and other mental health issues that can adversely affect students' well-being academic performance and overall quality of life. Traditional methods of mental health assessment and support are often resource-intensive, generalized, or inaccessible to many students limiting their effectiveness in providing timely and personalized interventions.

This project aims to address these challenges by leveraging survey data and predictive modeling to develop a system that identifies mental health risk levels among college students. By using machine learning algorithms the system will analyze factors such as stress, sleep quality, academic performance, and emotional resilience to classify individuals into risk categories. The ultimate goal of the project is to provide an efficient, accurate, and accessible tool to support

early detection and intervention in relation to mental health impairment and overall academic success.

1.3 Objective

This Project aims to address the challenges of mental health prediction through machine learning algorithms. Specific objectives include:

- Predicting Mental Health using Ml Models such as Random Forest, Logistic Regression, and Decision Tree to classify mental health as low, mid, and high.
- Comparing the performance of the models.
- User-friendly framework that does not require specialized hardware.

1.4 Scope of Study/Project

This project focuses on predicting mental health using machine learning algorithms. Key components of the research include:

Evaluating the performance of various machine learning algorithms. Developing a website accessible to students without requiring specialized devices. Work for deploying this in the educational sector.

By addressing these objectives, the research aims to contribute to advancement in the field of assistive technologies, ensuring timely and effective support for students.

Chapter 2

Literature Review

2.1 Literature Review

Many studies focus on the mental well-being of university students, exploring innovative approaches such as machine learning, mental health apps, and AI-driven solutions. Machine Learning-Based Prediction of Mental Well-Being Using Health Behavior Data from University Students (2023) applies machine learning algorithms, such as Random Forest, to predict mental well-being but is limited by its reliance on Southeast Asian data, which may not be generalizable. Future research could include diverse populations for more robust predictions. Prediction Modeling of Mental Well-Being Using Health Behavior Data of College Students (2022) predicts mental well-being but targets a specific population and recommends future research to develop models with more features. Experiences of Using Mental Health Apps to Support Psychological Health and Wellbeing (2023) studies MHApps and their use in distress relief, but its results may not generalize to all populations. More research is required to ascertain their applicability to a larger extent. Enhancing Mental Health with Artificial Intelligence: Current Trends and Future Prospects (2023) reviews AI in mental health, such as early detection and personalized treatments, but can bring no ultimate solution to all ethical concerns and require further development of regulatory frameworks. A Gamified App for Supporting Undergraduate Students' Mental Health: A Feasibility and Usability Study (2022) explored a gamified app using IoT to enhance mental well-being but with a very limited sample size, which would require future validation across diverse populations. Together, these studies highlighted the need for broader, more inclusive research and the integration of emerging technologies in supporting student mental health.

2.1.1 Key Theme and Areas Covered

- 1. Survey-Based Mental Health Analysis

Survey-based mental health research constitutes an unparalleled method to obtain a psychological portfolio of any individual, especially within the college milieu. It collates self-reported information on various dimensions of mental health, which include but are not limited to stress levels, anxiety, lifestyle habits, and coping mechanisms; thus helping make strides in understanding demand factors for mental health. The proposed application uses survey-based analysis to measure students' mental health needs and delivery of tailored interventions. Examples of such surveys are questions on emotional well-being, lifestyle preferences, school pressures, and social interactions. The data is subjected to processing and modeling using machine learning algorithms to find patterns, predict risks, and nurture actionable responses. Thus, the approach allows the identification of at-risk subjects and makes possible the intervention for them on a one-to-one basis. However, it will depend on efficiency on the quality and honesty of the responses. In union with advanced data analytics and machine learning, this method provides a scalable and effective approach to all other endeavors related to mental well-being in varying populations.

- 2. Machine Learning Algorithms for Mental Health Prediction

Machine learning is a very effective tool for predicting and intervening in the mental health of college students. It uses data from surveys, behavioral patterns, and usage data to identify mental health risks and develop personalized interventions. Logistic Regression is used for binary classifications, and Support Vector Machines (SVM) for high-dimensional classifications. The Random Forest algorithms determine key factors such as sleep quality and academic pressure, while K-Nearest Neighbors (KNN) groups students with similar patterns for targeted recommendations. These algorithms provide early warnings, actionable insights, and personalized support to enhance the effectiveness of mental health interventions. The choice of algorithm depends on the complexity of the data, the nature of the problem, and the goals of the project to ensure accurate and impactful solutions.

- 3. Dataset Development and Limitations

The dataset for mental well-being prediction includes a mixture of surveys, demographic information, behavioral records, and unstructured text. Accurate labeling done by trained mental health practitioners assures reliable labeling alongside diversity and confidentiality. Nevertheless, limitations exist such as data imbalance, self-survey data, limited access to real-time behavioral data from wearables, privacy issues, and ethics. Furthermore, data from a particular institution or region may not generalize well, which causes difficulty in applying the findings broadly. Overcoming such challenges, therefore, requires careful design, diversity of data collection, and attention to ethical principles.

- 4. Novel Approach to Accessibility

The purpose of the project is to create an inclusive website for the college-going population, removing mental health barriers like stigma and few resources. The app employs technology to develop a user-friendly design with simple navigation and customizable interfaces, accommodating users with varying levels of tech mastery or disabilities. It includes offline functions for areas where connectivity is low and multilingual support for really engaging purposes. The app allows for anonymous responses to address stigma and employs adaptive machine learning models to give personalized responses. Together, this means that the app is inclusive, anonymous, and adaptive, which extends its reach across a significant population base to promote mental wellness and lessen traditional barriers to care.

- 5. Integration of Prediction and Support Tools

This project aims to build up prediction and support tools that tackle the mental health needs of college students. The website relies on machine-learning models to predict the possible mental health risks by collecting information based on survey responses, behavioral patterns, and academic stress indexes. These predictions are then directed toward the development of custom-designed support interventions. The support tools offer personalized recommendations for mindfulness exercises, stress management tips, or counseling sessions. The app connects users with mental health professionals for credible support. Such an approach takes notice of the possible rising issues early and offers immediate,

evidence-based fixes thus improving the resilience and well-being of the students.

- 6. Early Prediction and Intervention

  The app uses machine learning models to identify early signs of mental health issues among college students. It uses data from surveys, behavioral patterns, and lifestyle indicators to provide personalized interventions, such as coping strategies and mindfulness exercises. This proactive approach enables students to get on top of managing their mental health, thus preventing potential adverse effects on academic performance and well-being. This proactive approach inspires resilience and promotes long-term mental health stability.

2.1.2 Search Strategy

The search strategy is directed at finding relevant data, research, and technological solutions to support the development of a mental health prediction and intervention tool for students. This method involves the following steps:

- Database Search: PubMed, IEEE Xplore, Scopus, SpringerLink, and Google Scholar.
- Keywords Used: Mental health in college students, Machine learning for mental health prediction, Digital mental health tools, Behavioral data and mental health, Stress and anxiety prediction.
- Inclusion Criteria: Peer-reviewed, college or young adult research on machine learning models and digital tools for mental health, as well as articles presenting these interventions.
- Exclusion Criteria: The study excluded studies involving clinical or medical treatments beyond digital tools, non-college-aged populations, non-peer-reviewed articles, insufficient data, or unclear methodologies.

2.1.3 Insights from Prior Work

Research indicates that mental health tools based on machine learning technology could help mitigate the barriers that digital platforms offer to students seeking mental health support. Stigma and a lack of time for in-person counseling can make it difficult for students to seek help. The success of the websites and online platforms retrain one

another to the respondent's mental health resources when they become available digitally in integrating into one's general lifestyle. Machine learning models like Random Forest, SVMs, and neural networks have shown promising results for identifying the at-risk individuals and predicting conditions like anxiety, stress, and depression. These models allow for timely interventions and personalized support tailored to each student's projections. However, issues such as the bias in machine-learning models, data privacy, and the generalizability of their findings in different student populations continue to affect the research. Connecting predictive models with personalized interventions in mindfulness exercises or links to professional support shows promise in improving the mental health outcomes of students.

Title & Year	Problem	Contributions	Limitations	Open problems/Future work
Machine Learning-Based Prediction of Mental Well-Being Using Health Behavior Data from University Students (2023)	University students face mental health challenges that affect their well-being and performance.	The study evaluates machine learning algorithms for predicting mental well-being, with Random Forest and boosting algorithms achieving the highest accuracy.	The study is based on self-reported data from a Southeast Asian student population, limiting generalizability.	Future research could include diverse populations and integrate more data sources for better predictions.
Prediction Modeling of Mental Well-Being Using Health Behavior Data of College Students (2022)	Lack of early identification and support for negative mental well-being in university students.	Developed machine learning models to predict mental well-being based on health behaviors and demographic data.	Focused on Southeast Asian university students; results may not be generalizable.	Further refinement of prediction models and exploring additional features for mental well-being monitoring.

Experiences of Using Mental Health Apps to Support Psychological Health and Wellbeing (2023)	Limited understanding of user experiences with mental health apps (MHApps) in supporting psychological health and well-being.	Explored user experiences with MHApps, highlighting their role in providing immediate distress relief distinct from traditional mental health services.	The study focused on a specific demographic; findings may not be generalizable to all populations.	Further research is needed to understand the broader applicability of MHApps across diverse user groups.
Enhancing Mental Health with Artificial Intelligence: Current Trends and Future Prospects (2023)	Mental health services often face challenges in accessibility, personalization, and efficiency.	The paper reviews AI applications in mental health, including early detection, personalized treatments, and AI-driven virtual therapists. It also discusses ethical considerations and future directions for AI in mental health.	The review may not cover all AI applications in mental health and may not fully address all ethical concerns.	Future research should focus on developing clear regulatory frameworks, validating AI models transparently, and conducting continuous research to enhance AI applications in mental health.
A Gamified App for Supporting Undergraduate Students' Mental Health: A Feasibility and Usability Study (2022)	Undergraduate students often face mental health challenges, and existing support systems may not effectively address their needs.	The study developed a gamified, theory-based mobile app that utilizes Internet of Things (IoT) technology to provide personalized data, aiming to enhance students' resilience and mental well-being.	The study's sample size was limited to 74 participants, and the findings may not be generalizable to all student populations.	Future research should empirically validate the app's design using quasi-experimental methods and explore its effectiveness across diverse student groups.

Table 2.1: Summary of Research on Mental Health Prediction

Chapter 3

Proposed Work

3.1 Proposed Work

The principal objective of the proposed work is to develop a website that leverages machine learning to predict and intervene in the mental health of college students. The application uses surveys, behavioral data, and performance indicators to identify students who may be at risk for developing mental health problems. It subsequently provides tailored interventions such as stress relief exercises and mindfulness activities to mitigate mental health challenges. The app will also connect students with professional counseling resources or hotlines. The end product is envisioned as a multilingual and friendly application accommodating all levels of technology proficiency and varying language preferences. Its vision is to develop a powerful tool for student mental wellness that manages risks early on in the process while providing students with tools to be proactive managers of their well-being. This mix of early risk indicators with support provides resilience to students, helping them through multiple challenges they face in college, thus positively impacting mental wellness and academic performance.

3.1.1 Objectives of the Proposed Work

The goal of this project is to create a website to uses predictive analytics and personalized interventions to take assistance on mental health among college students by identifying early signs of stress, anxiety, and depression, and providing solutions individualized to the user.

Primary Objective

The project is focused on creating a website using machine learning to predict mental health risks among college students and to offer students personalized interventions like coping strategies, mindfulness exercises, referrals to professional mental health resources, and support to improve student well-being and academic performance.

Secondary Objective

The app strives to enable accessibility, inclusivity, and ethical data handling through multilingual interfaces, customizable interfaces, and strong privacy measures and resilience through early support on arising mental health challenges and empowerment of students.

Third Objective

We have built a website where the user can take a survey and predict their mental health and If he has low mental health he can make appointments with our experties. If he has mid mental health then he will get recommendations to increase that by showing some videos regarding how to increase mental health and if he has high mental health he doesn't need to do anything.

3.2 Workflow

The system has three primary user roles: Admin, Consultant, and Student. Each role has distinct responsibilities and workflows designed to facilitate seamless interaction and service delivery.

3.2.1 Admin Role Workflow

The Admin plays a crucial role in managing platform operations, focusing on consultant registrations and platform monitoring.

- Review Consultant Registrations: The admin reviews new consultant registrations submitted through the platform.
- Verify Proof Documents: Verification of the uploaded documents is done to ensure the authenticity of consultants.
- Oversee Onboarding: Approved consultants are guided through the onboarding process.
- Monitor Platform Activities: The platform is regularly monitored to ensure compliance and efficiency.

3.2.2 Consultant Role Workflow

Consultants provide expert mental health services and engage with students.

- Registration Process: Initially the consultant has to register on the platform which will be approved by the Admin after verification.
- Upload Proof: Consultants have to submit the required documents for verification.

- Await Admin Verification: Consultants must await approval from Admin after document submission.
- Manage Bookings: After approval consultants can manage appointments and consultations.
- Engagement Activities: Consultants handle tasks such as marking cases as closed, sending messages to communicate with the students, and responding to community questions.

3.2.3 Student Role Workflow

Students can interact with the platform for mental health assessments, consulting experts, and community support.

- Student Login: Students can log in and can access the dashboard to navigate platform functionalities.
- Take Survey: Students complete a survey to assess their mental health status.
- Input Responses: Survey responses are entered into the system.
- Predict Mental Health Status: The system predicts the student's mental health status as
    - Low: Redirected to the homepage for general support.
    - Mid: Recommended suitable consultants for further assistance.
    - High: Directed to consult with experts urgently.
- Follow-Up Actions: Book a consultant, rate a consultant after receiving services, and message a consultant for ongoing discussions.

3.3 Methodology

The proposed mobile application will be developed through a systematic methodology, ensuring that each phase addresses the key objectives of early prediction, personalized intervention, and accessibility.

3.3.1 Overview of the Approach

Combining a mobile application that will incorporate predictive analytics, personalized interventions, and accessibility features will thus strengthen the mental health coping mechanisms of college students. In this manner, machine learning algorithms will analyze survey, behavioral, and academic performance data to predict mental health risks as early as possible. The app will offer recommendations such as Mindfulness Activities and Stress-Relief Activities specific to what a student may need. It will provide easy access to professional mental health support, such as counseling services and

hotlines. The app is to be easy to use, multilingual, customizable, and offline. Data privacy and ethical standards will be paramount in the development process.

3.3.2 Dataset Selection

This study focuses on the mental health of college students, with particular attention to stress, anxiety, wellness, and lifestyle habits. Other aspects may include demographic data, mental health indicators, lifestyle and behavioral factors, academic factors, emotional well-being, social and relationship factors, coping mechanisms, social anxiety, and emotional awareness. Information will also include previous experiences with a mental health diagnosis and treatment, such as counseling.

The survey will contain a wide range of self-reported questions regarding demographics, mental health indicators, lifestyle and behavioral factors, academic factors, emotional well-being, social and relationship factors, coping mechanisms, social anxiety, and emotional awareness. Data collection will be by anonymous surveys, thus ensuring confidentiality, ethical standards, and legal compliance. The overall purpose of the project is to provide essential information for machine learning models for predicting mental health risks and providing personalized interventions.

3.3.3 Algorithm Design

The algorithm design process for the prediction of mental health defines a systematic approach to handling survey data and predictions. Preprocessing data includes cleaning the dataset, normalizing numerical features, and encoding categorical responses into numerical representations. Feature selection techniques eliminate insignificant survey questions that wouldn't contribute towards the prediction of mental health. The data is then split into training and testing to assess the generalization of the model, and class-balancing techniques may also be used if necessary.

Model training involves the selection of an appropriate machine learning algorithm and tuning of hyperparameters. During the training phase, cross-validation is performed to ensure the robustness of the model on various data folds. Predictions are made from the trained model, and prediction outcomes are classified into filled-in mental health risk levels. Evaluations of the model predictions will include accuracy, precision, recall, and F1 score measurement and confusion matrices. In an application interface, the trained model may be implemented in which users are allowed to enter survey responses and receive friendly, in-time assessments of mental health risks. Such an organized approach guarantees correct predictions and actionable further decay assistance needed to cascade timely interventions beneficial for the improvement of mental health.

3.3.4 Tools and Technologies

The project utilizes Python for prediction and HTML, CSS, and JS for frontend development, And for backend we have used PHP. Supported by libraries and Flask for integrating the backend with ML. Development tools like VS code, Git, Google Collab, and XAMPP are employed to streamline the coding and collaboration process. We are hosting the website in XAMPP because it provides Apache for hosting and MySQL for the database.

3.3.5 System Architecture/Design

The survey-based mental health prediction model works via a structured survey collecting information on mental health, lifestyle habits, and emotional well-being. A Random Forest machine learning model processes this data on unacceptable risks whilst predicting the occurrence rate of mental illness. If everything reported is okay, it concludes. If anything wrong is detected, risk will be classified according to the level of prediction: Low, Mid, or High. This machine learning automated protocol allows the attribution of insights for possible interventions and gives an overall enhancement in mental health.

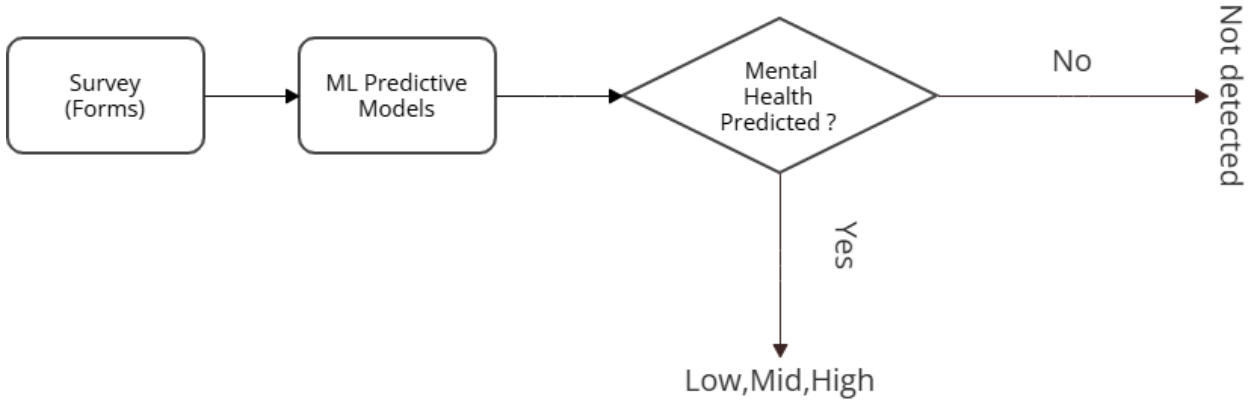

Fig 3.1: System Architecture for Mental Health Prediction

3.3.6 Algorithm Description

The mental health prediction algorithm will follow various steps to preprocess survey data. Survey data from users contain questions about their mental health, lifestyle, or emotional well-being, forming the input dataset. Data preprocessing prepares this previous dataset for some machine learning models. Such a model analyses the data to find possible patterns and correlations to classify individuals as low, medium, or highly

risk. In the case of an insignificant risk factor, however, a "Not Detected" result will be outputted by the system. Because the decision-making is based on the learned patterns, accurate predictions are assured. This architecture incorporates survey data collection, machine learning-based analysis, and actionable outputs. Predictions would lead to an almost immediate identification of mental health risks, allowing the users to seek the correct support or intervention. The system emphasizes efficiency, accuracy, and user-friendliness, imparting great value in making it worth the solution to mental health issues.

3.3.7 Expected Outcomes

Through machine learning models, the mobile application develops early mental health support for college students by identifying signs of stress, anxiety, and depression. Early identification will pave the way for timely interventions, providing users a better chance at prevention and enhancing general mental well-being. The app will also provide personalized mental health recommendations, mindfulness exercises, and stress-relief activities as some examples to the user in ways that promote their mental health management. The app will also direct users to obtain help from professional support services, which may include counseling and therapy. The app promises to improve mental health outcomes while boosting academic performance and resilience among students. The user-friendly design, its accessibility features, and data-driven insights will ensure all-encompassing guidance in handling any kind of mental health problem.

3.3.8 Advantages of the Proposed Work

The mobile application is known for various benefits for college students concerning mental health. Using predictive machine learning models, it detects possible mental health risks early, allowing early intervention where students have the potential to develop conditions like stress, anxiety, and depression. Besides, the application uses your personal data to provide individualized mental health recommendations that include mindfulness exercises, stress-relief activities, and coping strategies. These recommendations are dynamically tailored continuously, based on the needs of different possible students. Accessibility and usability are achieved through an intuitive interface, multilingual capabilities, and offline performance. Finally, the app features a unique integration to access all mental health professionals, including therapy and counseling services. These facilities, taken collectively, shape the application to be quite comprehensive, scalable, and pragmatic to tackle mental health challenges in a preemptive yet consequential manner.

3.3.9 Limitations and Assumptions

The proposed mobile application for mental health support enjoys several limitations, such as the use of subjective self-reporting data that can be susceptible to biases and

inaccuracies; rather, a proper selection of samples can affect the efficiency of machine learning models. Other limitations include poor user engagement and adherence, as well as technical limitations, such as Internet connectivity, compatibility with older devices, and the like, which can hinder its adoption. The app presupposes that students would be inclined to share their personal data and would trust the app's privacy and security provisions. It assumes also that machine learning models can actually predict mental health risks accurately based on the dataset in question, even though the intended dataset might not tell the whole story. It also presupposes that improving patient outcomes would be associated with personalized recommendations and courses of action leading to professional help connecting with the user. However, these limitations also represent opportunities for iterative development, algorithmic refinement, and user-centered design to increase the reliability and effectiveness of the app over time.

Chapter4

Experimentation and Results

4.1 Experimental Setup

All experiments were conducted on a standardized computational setup to ensure consistency across the evaluated algorithms. The hardware environment consisted of an Intel Core i5-1135G7 processor, NVIDIA GeForce MX350 GPU with 6GB VRAM, 8GB DDR4 RAM, and a 256GB NVMe SSD. The system ran on Windows 11.

4.2 Dataset

The dataset used for evaluating the effectiveness of our project consists of 83 entries. Participants were S7 B.tech CSE students. The data were given to the algorithms. Labels were determined by experts, focusing on the scales that the participants have given in the survey. This dataset serves as a benchmark to predict mental health to distinguish between high, mid, & low.

4.3 Evaluation Metrics

Various metrics, such as prediction Accuracy, precision, Recall, and F1 score will be used to assess the effectiveness and performance of the proposed mobile application for mental health support.

Accuracy measures the proportion of correct predictions made by the machine learning model. It is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP is the no: of true positives,
- TN is the no: of true negatives,
- FP is the no: of false positives, and
- FN is the no: of false negatives.

Precision measures the proportion of correctly predicted positive cases out of all cases predicted as positive. It is the ratio of true positive predictions to the total number of instances predicted as positive. It is calculated as:

$$Precision = \frac{TP}{TP + FP}$$

A higher precision value indicates that the model has a low false positive rate.

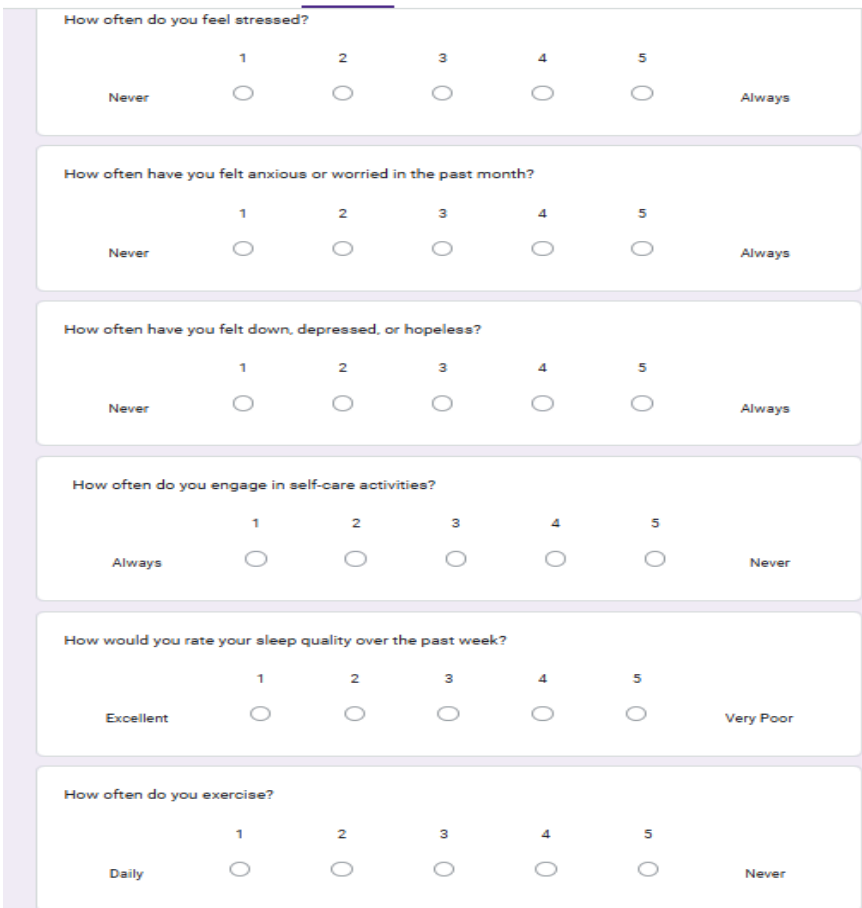

Fig 4.1: Dataset Collection Format

Recall measures the proportion of correctly predicted positive cases out of all actual positive cases. It is the ratio of true positive predictions to the total number of actual positive predictions to the total number of actual positive instances. It is calculated as:

$$Recall = \frac{TP}{TP + FN}$$

A high recall value indicates that the algorithm is identifying most of the true positive cases, though it may also include some false positives.

F1 score is the harmonic mean of precision and recall, providing a balanced measure that considers both false positives and false negatives. It is particularly useful when there is an imbalance between the classes, as it considers both false positives and false negatives. The F1 score is computed as:

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

4.4 Experimental Design

The proposed mobile application aims to evaluate how effectively it could predict mental health risks, offer personalized recommendations, and improve user well-being. It includes dataset collection, system testing, and user engagement analysis. The experiments focused on predicting on a standardized dataset of 83 data partitioned into training, validation, and testing sets with an ratio. The implementation is conducted on a system Intel Core i5-1135G7 processor, NVIDIA GeForce MX350 GPU. Training protocols involved testing learning rates (), standardized batch sizes, and varying epochs to ensure convergence. Performance evaluation metrics included accuracy and training time to ensure robustness. This experimental design ensures a comprehensive evaluation of the application's functionality, predictive accuracy, and impact on students' mental health, guiding further enhancements for real-world deployment.

4.4.1 Experimental Scenarios

The proposed mental health app shall be subjected to a variety of experimental scenarios. The first is data collection and preprocessing, whereby students will fill out surveys focusing on their mental health, lifestyle, and academic stress. The second scenario will evaluate the predictive models Regression, Random Forest, and SVM for effectiveness. Experimental setup three involves assessing the user-specific personalization system, which generates customized interventions based on simulated user profiles. Experimental scenario four centers on user engagement and retention, measuring the frequency of sessions, their duration, and adherence to suggestions. Experiment five will determine the impact of the application on mental health measures by using pre- and post-study questionnaires to compare self-reported levels of stress, anxiety, and improvements in behavior. Lastly, scenario six will examine scalability and performance via a concurrent usage simulation, and stress test the backend against real-time prediction and recommendation delivery loads. All scenarios are collectively designed to rigorously prepare and perfect the application for real-life deployment.

4.4.2 Parameter Tuning

Parameter tuning is a vital step for optimizing the performance of machine learning models toward mental health applications by adjusting the hyperparameters for better accuracy, precision, recall, and F1 value. In models such as Logistic Regression, hyperparameters are the regularization parameters chosen to balance bias against variance. In Support Vector Machines, selected parameter types are regularization and the choice of kernel, the objective being to adjust decision boundaries. A Random Forest model requires tuning the number of trees, maximum depth, and minimum sample split, while K-Nearest Neighbors needs to tune the number of adjacent samples and distance metric. Techniques for parameter tuning may include cross-validation and

automated techniques, like Bayesian optimization or grid search. The results of parameter tuning are validated on a separate test set to confirm the improved predictive performance of the model and also increased reliability in mental health risk assessment.

4.5 Results

The performance of machine learning models for predicting mental health uses the Random Forest algorithm to an effective degree compared to the Decision Tree model. The accuracy rates of 0.98 and 0.92 in both training and testing locations show the Random Forest model's high potential to generalize well and run efficiently on data never seen before. This impressive performance demonstrates the strength of Random Forest's ability to handle unevenly complicated and variable datasets while minimizing the likelihood of overfitting through its ensemble technique.

On the contrary, Decision Tree recorded a training accuracy percentage of 0.93, which is a good but not perfect result. Conversely, its accuracy is diminished to 0.76 when measuring the unseen data tested, possibly indicating an overfitted model with low generalizability. This difference itself suggests the shortcomings of single-tree models in capturing complex patterns and the tendency towards overfitting, especially with smaller or unbalanced datasets.

The results evidently highlighted the better fit for the project by Random Forest, providing improved accuracy and reliability in predicting risks to mental health among college students. The ensemble-based model secures increased stability and correctness and becomes a strengthened solution in providing practical insights and recommendations.

4.6 Analysis of Results

4.6.1 Random Forest Model

The Random Forest model is well poised to predict mental health outcomes with a training accuracy of 0.98 and a testing accuracy of 0.92. The model is effective in dealing with complex data and provides significant accurate predictions. The Random Forest, as an ensemble model that makes use of many decision trees with the aim of reducing issues of overfitting, is also rather stable. Capability of feature importance helps underline some of the important components that influence mental health prediction such as stress levels, quality of sleep, or academic pressure. Access to this information becomes critical in guiding interventions to such a target that some challenges will be met. The model is able to classify mental health at risk into Low, Medium, or High, developing further into an actionable suggestion. To prove the accuracy of the model with respect to prediction low false positives and negatives were

obtained by the use of salient metrics such as precision, recall, and F1 score. In conclusion, Random Forest can operate as a powerful model for predicting mental health outcomes and thus aid in intervention subtly aimed.

4.6.2 Decision Tree

The Decision Tree model in predicting mental health has shown a Training Accuracy of 0.93 and a Testing Accuracy of 0.76. This means that while the model learns the patterns of the training dataset very well, it less effectively generalizes to unseen data. This indicates potential overfitting, as the model becomes too closely tailored to the training data and therefore fails to generalize well to new data. Decision Trees also provide transparency in the modeling process. They give a visual representation of the splits and decisions used to classify a given level of risk for mental health, thereby aiding in understanding the effect of individual features used to make predictions. Nevertheless, the lower testing accuracy only points to the general fact that the model is sensitive to noisy data and thus needs a more complex approach. Some of the techniques that can help include pruning or the possibility of its use in tandem with ensemble methods for increases in generalization. Overall, the Decision Tree model is still a very basic method that needs to be carefully tuned or more methods combined with it that will bring better suitability with the real world.

4.7 Observations

4.7.1 Random Forest Model

The random forest model is performing wonderfully with very high training and testing accuracies of 0.98 and 0.92 respectively. The decrease in accuracy from training to testing is marginal, which points out the strength in generalization over unseen data-a major advantage with random forests. It, therefore, indicates that it has learned the underlying patterns in images and has not over-fitted. However, it still holds a capacity for improvement- hyper-parameter tuning or maybe getting the class imbalance in further accord would achieve better performance.

4.7.2 Decision Tree Model

On the contrary, the decision tree model shows a training accuracy of 0.93 which is good but less than that of the random forest model. The drop in testing accuracy is very steep, landing at 0.76. This shows there is a problem with generalization. The gap between training and testing accuracy is quite large, and this might be due to overfitting, in which the decision tree identifies noise or unnecessary detail in the training data. To

improve this model, techniques like pruning could be employed to reduce overfitting, or an ensemble method like random forest could be used to boost its generalization capabilities.

4.7.3 Dataset Constraints

The random forest and decision tree models, both with a small sample size of 83 data points, face challenges due to their sensitivity to the small dataset. The random forest model can reduce overfitting by averaging predictions from multiple trees but may overfit if not optimized. The decision tree model is more likely to overfit due to its ability to capture noise or irrelevant patterns. Pruning or setting depth limits can help mitigate overfitting. Both models may struggle with generalization, and obtaining more data, cross-validation, and regularization techniques are essential steps to improve their performance.

4.7.4 Algorithms Efficiency

Algorithms like random forest and decision trees vary with factors like dataset size, model complexity, and computational power. Decision tree models are usually very fast to train on smaller datasets, but very deep or complex trees can dramatically slow them down. A good pruned decision tree will, however, have good performance versus computation.

On the other hand, random forest algorithms are more computationally expensive since they require training multiple trees and aggregating their predictions. Although they are more robust and less prone to overfitting, their efficiency can decrease with the growth of the dataset size and number of trees. However, for a small dataset, random forest may not be as computationally intensive as larger datasets.

Decision trees are usually faster and simpler in terms of algorithm efficiency than random forests, especially with small datasets. However, they offer better generalization and accuracy at the cost of more computational resources.

4.7.5 Model Comparison

The comparison of machine learning models for mental health prediction highlights their strengths and weaknesses in terms of training and testing accuracy. Here is a detailed evaluation of the models used:

Model	Training Accuracy	Testing Accuracy	Strengths	Weaknesses

Random Forest	0.98	0.92	High accuracy, robust to overfitting, interpretable feature importance.	Slightly lower generalization compared to ensembles.
Logistic Regression	0.98	0.8	Simplicity, fast computation, good for linear relationships.	Moderate generalization for complex datasets.
Decision Tree	0.93	0.76	Transparent decision-making, interpretable.	Prone to overfitting, lower testing accuracy.
SVM	1	0.56	Effective in high-dimensional spaces.	Severe overfitting, poor generalization.
XGBoost	1	0.88	High accuracy, efficient handling of large datasets.	Slight overfitting to training data.
K-Nearest Neighbors	0.82	0.72	Simple implementation, effective for small datasets.	Limited scalability, lower testing accuracy.
Voting Classifier	1	0.84	Combines strengths of multiple models, robust performance.	Slight overfitting, computationally expensive.
Naive Bayes	0.84	0.76	Fast computation, suitable for categorical data.	Assumes feature independence, and moderate accuracy.

Table 4.1: Model Comparison

4.8 Comparative Analysis

The random forest model outperforms the decision tree model in terms of training accuracy and testing accuracy, with a training accuracy of 0.98 and a testing accuracy of 0.92. This is because the random forest uses an ensemble approach, which reduces variance by averaging predictions across multiple trees. However, the decision tree is more prone to overfitting, especially when the dataset is small, as it can easily capture noise or irrelevant patterns.

The decision tree, therefore, has the advantage of being faster and less resource-intensive for computation, suitable for smaller datasets, or when the efficiency of the computation is very critical. On the other hand, the random forest requires more resources because it builds multiple trees, which could slow down the training and prediction times. However, because the dataset is very small at 83 data points, the added computational cost may be minimal.

The decision tree is more interpretable because it provides a structure that can be visualized and understood. The random forest, being an ensemble of many trees, is less interpretable, as it combines the outputs of multiple models, making it difficult to pinpoint the exact decision-making process. However, when the accuracy is higher, as in the case of the random forest model, then the loss in interpretability does not matter much.

In summary, the random forest model is less prone to overfitting and, thus more robust and accurate but at the cost of extra computational resources and loss of interpretability. The decision tree gives a simpler and more efficient model but struggles to generalize quite nicely on small datasets.

4.8.1 Limitations of the Proposed Architectures

The random forest and decision tree architectures have their strengths but also their limitations. The random forest model is computationally expensive, which results in slower training and prediction times, especially when dealing with large datasets or resource-constrained environments. Its lack of interpretability makes it harder to understand and explain the decision-making process compared to simpler models, which can be a drawback in domains where model transparency is critical.

Decision trees are likely to overfit and become unmanageable, hence difficult to interpret, with large depths of decision trees and smaller datasets that cause a reduced generalization. The decision tree is highly susceptible to data quality and noise in the datasets can bias decision trees. These models are efficient from a computation point of view but fail to address complex relations that may be identified by a more advanced model.

Both models suffer in handling high-dimensional data. Decision trees might be unable to manage large feature sets without proper pruning, and as the number of features grows, random forests could see diminishing returns. In a nutshell, both architectures have their merits but each at a cost in terms of computational complexity, interpretability, and noisiness or complexity of data.

Chapter 5

Conclusion and Scope for Future Research

The proposed mobile application effectively tackles a key interest in the provision of mental health support for college students using machine learning and personalized advice. The results suggest that it can accurately assess a student's risk of mental health concerns, make personalized interventions, and favorably affect the user's state of well-being. The application combines surveys, machine learning algorithms, and user feedback in real time, allowing students to gain tools for the management of stress, anxiety, and related problems. The pilot study did show that it is effective, yielding significant improvements in both mental health measures reported by users and their behavioral metrics. This finding opens avenues for establishing the importance of tech-driven solutions to improve mental health awareness and resilience at an academic level.

Though promising, these results indicate that very large untouched areas of future research and development await the growth of this project. In particular, broadening the dataset will include a more diverse set of participants, thereby improving the generalisability of predictions. Adoption of advanced techniques like deep Learning in mental health assessments and recommendations could increase accuracy. There is a need for longitudinal evaluation of established mental well-being profiling with growing use in the management of mental health. Besides, adding extra features in the app, peer support networks, gamification, and multilingual ability should be implemented to enhance user engagement and accessibility.

Scope for Future Research

- Dataset Expansion: Collect a large amount of data from people of different age groups.
- Feature Exploration: Adding new features for the users.
- Model Enhancing: Increasing the model accuracy in predicting mental health.

This research plays a significant step in predicting mental health and giving recommendations to increase your mental health.

Appendix A

Source code

FINAL.ipynb

File Edit View Insert Runtime Tools Help Changes will not be saved

```python
import pandas as pd
df = pd.read_csv('Updated_Mind_Sync_Mental_Health_Survey.csv')
df.columns = df.columns.str.strip()


numerical_columns = df.select_dtypes(include=['int64', 'float64'].columns
for col in numerical_columns:
    df[col] = df[col].fillna(df[col].mean())

categorical_columns = df.select_dtypes(include=['object']).columns
for col in categorical_columns:
    df[col] = df[col].fillna(df[col].mode()[0])

unnecessary_columns = ['Timestamp']
df = df.drop(columns=unnecessary_columns, errors='ignore')


gender_mapping = {
    'Female': 0,
    'Male': 1,
    'Other': 2,
    'Prefer not to say': 3
}
df["gender"] = df["gender"].map(gender_mapping)

diagnosed_mapping = {
    'No': 0,
    'Yes': 1
}
```

FINAL.ipynb

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```python
df["Have you ever been diagnosed with a mental health condition?"] = df[
    'Have you ever been diagnosed with a mental health condition?'
].map(diagnosed_mapping)

therapy_mapping = {
    'No': 0,
    'Yes': 1
}
df["Have you ever attended therapy or counseling sessions?"] = df[
    'Have you ever attended therapy or counseling sessions?'
].map(therapy_mapping)

risk_levels_mapping = {
    'High': 0,
    'Low': 1,
    'Mid': 2
}
df["Risk Levels"] = df["Risk Levels"].map(risk_levels_mapping)

df.to_csv('Preprocessed_Mind_Sync_Mental_Health_Survey.csv', index=False)

print("Data preprocessing complete. Preprocessed dataset saved as 'Preprocessed_Mind_Sync_Mental_Health_Survey.csv'.")
```

```
Data preprocessing complete. Preprocessed dataset saved as 'Preprocessed_Mind_Sync_Mental_Health_Survey.csv'.
```

Fig A.1: Preprocessing

FINAL.ipynb

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```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.impute import SimpleImputer
from sklearn.model_selection import GridSearchCV

df = pd.read_csv('Preprocessed_Mind_Sync_Mental_Health_Survey.csv')

X = df.drop(columns=["Risk Levels"])
y = df["Risk Levels"]

imputer = SimpleImputer(strategy='mean')
X_imputed = imputer.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_imputed, y, test_size=0.3, random_state=42, stratify=y)

random_forest = RandomForestClassifier(random_state=42)

param_grid = {
    'n_estimators': [50, 100],
    'max_depth': [10, 20, None],
    'min_samples_split': [2, 5],
    'min_samples_leaf': [1, 2],
    'criterion': ['gini']
}

grid_search = GridSearchCV(
    random_forest, param_grid, cv=3, scoring='accuracy', n_jobs=-1
```

FINAL.ipynb

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```python
)
grid_search.fit(X_train, y_train)

best_model = grid_search.best_estimator_

train_predictions = best_model.predict(X_train)
test_predictions = best_model.predict(X_test)

train_accuracy = accuracy_score(y_train, train_predictions)
test_accuracy = accuracy_score(y_test, test_predictions)

print(f"Training Accuracy: {train_accuracy:.2f}")
print(f"Testing Accuracy: {test_accuracy:.2f}")
```

```
Training Accuracy: 0.98
Testing Accuracy: 0.92
```

```python
new_datamid = np.array([
    [21, 1.0, 3, 2, 3, 3, 1, 3, 1, 3, 2.0, 3, 3, 3, 2.0, 2, 3, 3.0, 3.0, 1, 3.5357142857142856, 3, 4, 0.0, 0.0, 3]
])
```

```python
new_datahigh = np.array([
    [20, 1.0, 4, 5, 3, 3, 2, 4, 4, 3, 5.0, 5, 4, 4, 3.0, 3, 1, 3.0, 4.0, 1, 3.5357142857142856, 4, 5, 0.0, 0.0, 3]
])
```

```python
new_datalow = np.array([
    [20, 1.0, 3, 2, 1, 1, 2, 3, 2, 3, 2.0, 2, 2, 2, 2.0, 2, 1, 2.0, 2.0, 1, 3.5357142857142856, 1, 1, 0.0, 0.0, 2]
])
```

Fig A.2: Training Random Forest Model

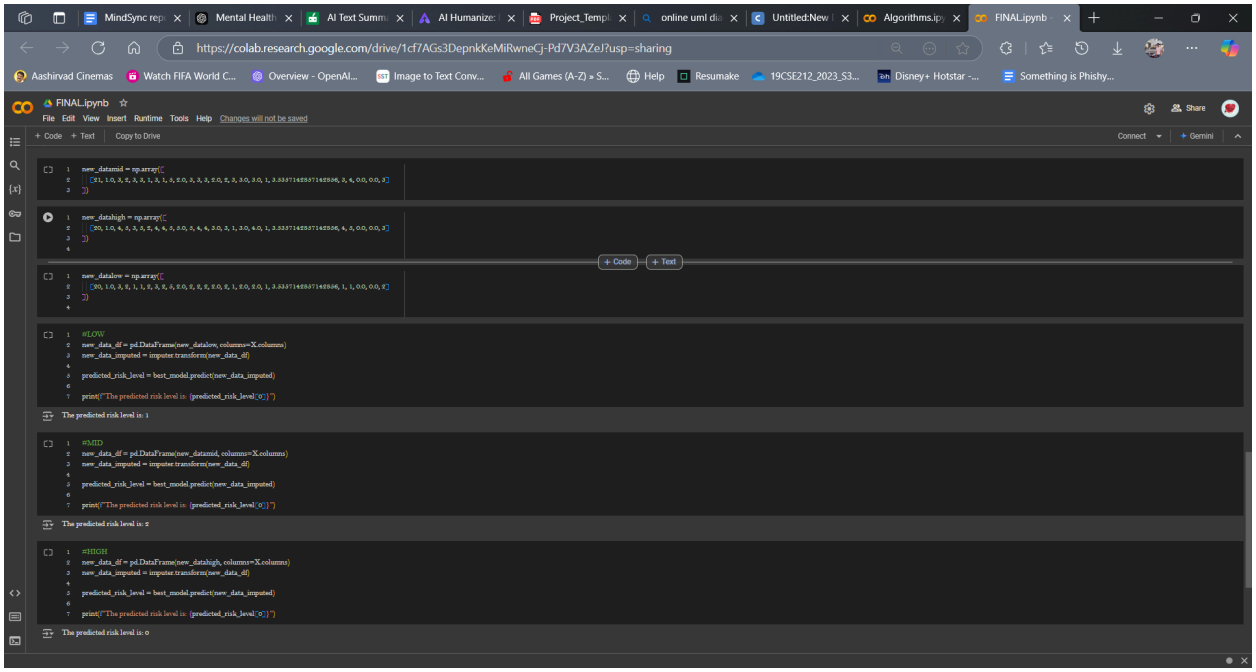

Fig A.3: Prediction

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